CURBCOUNT · RISK REGISTER AND HAZARD ANALYSIS

CurbCount risk register and hazard analysis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	Opened in the kit 1.9.1 upgrade: project risks, preliminary hazard analysis and first-pass design FMEA

CurbCount risk register and hazard analysis

The top risks are that the counts will not meet the accuracy targets on warm days (K1, F8: the highest RPN, 64), that charging stops at cold sites (K2) and that a part falls from a pole (H1, mitigated by bands, spacers and lanyards). Method: STANDARDS section 23. Likelihood, impact, severity, occurrence and detection are scored 1 (low) to 5 (high). Nothing here has been tested; every control is a design intent checked on paper.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
K1	Counts miss the accuracy targets	Cars cross the field in 4.2 frames, random merging loses 6.5 %, sunlit pavement hides people (CAL-001 C, D)	The counter fails PR-001, PR-002 or PR-008 in the field	3	4	Design lead	Track averaging in the tracker; compare with manual counts at the first partner street (TRL 4, on hold); radar variant for hot sites
K2	Node stops at cold sites	LiFePO4 may not be charged below 0 degrees C; charging stops at about 0 degrees C air or below (STM S8)	Counters at cold sites run dry after about 4 days of cold	3	3	Amish; FieldNode project	Decision O1 in CBC-DEC-001: heated cell pocket, low-temperature cell, or limit the claim to cool sites (recommended now)
K3	Parts cost passes the target	BOM is 1.00 USD under 275 USD on indicative prices	The target is missed by the first real quote	4	2	Design lead	Check prices when parts are bought; the target is a control figure, not a limit; savings listed in CBC-DEC-001
K4	No partner street with manual counts	No partner agreed; Toronto is only a candidate (CBC-DDR-002 O1)	Accuracy cannot be checked	3	3	Amish	Approach the first candidate; keep TRL 4 on hold until agreed
K5	FieldNode core changes under CurbCount	Pin order, port supply and jumper sensing are agreed with FieldNode but not final	The sensor port and BOM lines 1, 2, 11 and 14 change	3	3	Design lead; FieldNode project	Keep the agreement items in CBC-DEC-001; re-run the trace and the BOM when FieldNode freezes
K6	Residents distrust a thermal counter	A thermal camera is still a camera to many people	The pilot is refused or the device removed	2	4	Amish; partner city	Public notice plate with an accurate sentence (CBC-DDR-001 D7), open firmware, 7.6 px/m published, no frame leaves the device
K7	Window film fails in sun	Film grade unchosen; unstabilised polyethylene embrittles (STM S18)	The array loses transmission or the head leaks	3	3	Design lead	Specify a UV-stabilised clear grade; replaceable by design (clamped frame)
K8	Radar option cannot be shown to fit	Module price unknown, 60 GHz rules to assess (REG G9)	Option B drops out as the hot-climate variant	3	3	Amish	Research module price and rules before choosing B for any site
K9				3	2	Coordinator	

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
	Kit or render pipeline gaps delay the media	Renders are made on Amish's Mac, not here; media/options.png is not in this commit	Documents and renders are out of step				Render and compose the options picture after this commit

Preliminary hazard analysis

Table 2. Hazards. The counter has no mains connection and no moving parts; its hazards come from working at height, a falling part, a lithium cell and the privacy promise.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Panel or head falls onto the street	Band slips, bolt or set screw loosens, 35 m/s gust	A person below is hit	5	2	Four stainless band clamps at 1,000 N preload give twist factor 3.3 (CAL-001 H3); crush spacers; nyloc nuts; panel mount and head each on a lanyard (PR-021)	Low
H2	Fitter falls or drops a tool at height	Work at 4 to 5 m on a mobile platform	Injury to the fitter or a passer-by	4	2	Two people; panel fitted on the ground first; head already on the arm; work-at-height rules of the owner (REG G8, G11)	Medium until a partner's procedure is known
H3	Contact with pole or street-light wiring	Fitter opens or touches pole electrics	Electric shock	5	1	No drilling, no pole wiring, clamp-on only (PR-014); the counter has no mains connection; the pole owner isolates if asked	Low
H4	Lithium cell fault	Short circuit, over-temperature, charging below 0 degrees C	Heat, venting, fire in the enclosure	4	2	Fused holder 5 A, NTC lockout and protection on the FieldNode board (BOM lines 2 and 3); cell kept in the IP65 enclosure	Low
H5	Cut metal edges	Sawn and drilled aluminium and stainless	Cuts to the builder	2	3	Edge break and deburr on every cut edge (modelled with `break_edges` in CAD); gloves in BLD-001	Low
H6	Wind loads the pole	Panel and head act as sails	Pole or fixing overloaded	4	1	Panel 52 N and head 8 N in a 35 m/s gust; post factor 55 (CAL-001 H1)	Low
H7	Counter turned into a camera	Someone changes the firmware to store or send frames	Loss of the privacy promise	4	2	Open firmware and hardware, no code path that writes or sends a frame (STM S20), frames only on a service jumper, notice plate; a hardware change is visible	Medium; audit at TRL 4
H8	Head struck by a high vehicle	Head 4.3 m above the road	Head torn off and dropped	3	2	Head on a lanyard (PR-021)	Low to medium
H9	RF exposure under the radar (option B)	60 GHz transmitter 2.55 m above a person's head	Exposure above the guideline	2	1	12 mW/m2 on a 1 W overestimate against 10 W/m2 (STM S21)	Low

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection (maximum 125). Actions are required at an RPN of 36 or more and are proposals for the design review.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Band clamps	Hold the saddle plates to the pole	Loosen or slip	Arm twists, aim lost, parts fall	Preload lost by vibration or creep	5	2	3	30	Fit to the band torque that gives 1,000 N preload (to confirm, CBC-DEC-001 item 5); V-saddle grip; lanyards; inspect on each visit
F2	Pole-top sleeve and set screws	Hold the panel	Set screw loosens	Panel rotates or lifts	Vibration	4	2	3	24	Thread-locking compound; lanyard
F3	Arm bracket bolts and crush spacers	Carry the head on the arm	Bolt loosens or spacer crushes	Arm sags, aim drifts	Torque too high or too low	4	2	2	16	Spacers stop over-tightening (modelled); nyloc nuts
F4	Printed ASA head housing	Hold the array and keep out water	UV or heat cracking	Water in, array lost	Wrong print settings, long sun	3	2	4	24	Print in ASA, a UV-stable class (BOM line 8); inspect after the first summer
F5	HDPE window film	Pass 8 to 14 um radiation and seal the head	Embrittles, scratches or tears	Transmission falls, ingress	UV, grit, wrong grade (STM S18)	3	4	3	36	Specify a UV-stabilised clear grade; clamped frame so it can be replaced
F6	Window frame gasket	Seal the head to IP65	Gasket under-compressed	Water enters	Screws too loose, ledge uneven	4	2	3	24	Tighten the frame screws evenly (BLD-001); check by spray test at TRL 4
F7	M12 cable and M16 gland	Carry I2C and supply 2 m to the head	Gland leaks or cable chafes	Moisture, bus errors	Gland not tightened, sharp edge	3	2	2	12	Moulded plug, 6 mm cable in a rated gland; lanyard and ties
F8	Thermal array and window	Sense people against pavement	Contrast too low on sunlit pavement	Pedestrians missed for hours	Warm surfaces in sun (STM S3)	4	4	4	64	Track averaging; measure contrast at TRL 4; radar variant for hot sites
F9	Tracker on the STM32WL	Turn blobs into counts	Merges, splits or loses tracks	Wrong counts	Groups, occlusion, 4.2 frames per car	4	3	4	48	Tracker profile and replay on recorded frames at TRL 4; ESP32-S3 fallback head costs about 311 USD

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F10	LiFePO4 cell and charger	Store energy	Cell cannot charge in cold	Node stops after the stored days	Air below about 0 degrees C (STM S8)	3	3	2	18	Decision O1; fuel gauge reports state of charge
F11	Solar panel	Harvest energy	Shading or dirt	Low harvest, flat cell	Trees, dust, bird droppings (40 % derating assumed)	2	3	3	18	Derating in CAL-001 E; panel tilt and site check
F12	Enclosure lugs and saddle plate	Clamp the FieldNode enclosure to the pole	Lug screw loosens	Enclosure slips or tips	Vibration	5	2	3	30	Nyloc nuts on M5 lug bolts; bands carry the load, not the lugs
F13	Aluminium against stainless and a steel pole	Carry the load for five years	Galvanic corrosion at joints	Joint weakens or seizes	Wet, salted street (STM S19)	3	3	4	36	Coat or isolate contact faces; stainless against aluminium kept to small areas
F14	Calibration jumper and frame stream	Allow an aim check on site	Left connected	Frames stay viewable on the service port	Fitter forgets	4	2	3	24	Firmware sketch streams only while the jumper is present; close-up step in BLD-001
F15	Public notice plate	Tell the public what is counted	Ties fail or print fades	Plate lost or unreadable	UV, weather	2	3	2	12	Two stainless ties; UV-stable print
F16	LoRa antenna and uplink	Send the 10-byte record	Detuned by metal near the antenna	Lost packets	Pole and enclosure placement	2	3	3	18	Antenna on the enclosure top, clear of the arm; link test on site
F17	Option B radar bursts	Give the tracker 4 frames per vehicle at 0.2 W	Burst rate too low	Missed vehicles	Energy fix (CBC-DEC-001 C1) cuts frames	3	3	3	27	Profile the burst rate with the tracker (TRL 4)

Highest RPN: F8 (64), F9 (48), F5 (36). Detection scores assume no field testing yet (TRL 4 is on hold), so they are conservative.